# 14 · Karyotypes and aneuploidy

The words of this track, as it says them. Bold brown is the examiner.

**Karyotypes and aneuploidy. Four marks.**

**What is a human karyotype, and how many chromosomes does it show? What is aneuploidy? Give two examples, and explain how aneuploidy arises.**

The hook: forty-six is twenty-three pairs. One too many, or one too few, is aneuploidy, and it comes from a pair failing to split in meiosis.

A karyotype is a picture of a cell's chromosomes, taken at metaphase, arranged into numbered homologous pairs, by decreasing size. It's the complete diploid set.

Humans have forty-six chromosomes, which is twenty-three pairs: twenty-two pairs of autosomes, and one pair of sex chromosomes. X X in a female, X Y in a male.

Aneuploidy is an abnormal number of chromosomes in a cell. Too many, or too few.

Two examples. Down syndrome, which is trisomy twenty-one: three copies of chromosome twenty-one, so forty-seven chromosomes. And Turner syndrome: only one X, so forty-five. Edwards syndrome, trisomy eighteen, is another.

How it arises: non-disjunction. A pair of chromosomes, or its chromatids, fails to separate in anaphase of meiosis, one or two. So a gamete gets one chromosome too many, or one too few, and so does the zygote it makes.

The traps. Forty-six chromosomes is twenty-three pairs: twenty-two pairs of autosomes, plus one pair of sex chromosomes. "Twenty-three autosomes" is wrong. Second: Turner is forty-five, Down is forty-seven. Third: non-disjunction happens in meiosis, when the gametes are made.

**Once more. Karyotypes and aneuploidy.**

A karyotype is a picture of the chromosomes in numbered homologous pairs: forty-six chromosomes, twenty-three pairs, twenty-two pairs of autosomes plus X X or X Y. Aneuploidy is an abnormal number of chromosomes. Down syndrome is trisomy twenty-one, forty-seven chromosomes. Turner syndrome is forty-five, with only one X. It arises from non-disjunction in meiosis: a pair fails to separate, so a gamete gets one too many, or one too few.